Q1.

The drawing below shows a mole. Moles dig tunnels through soil.



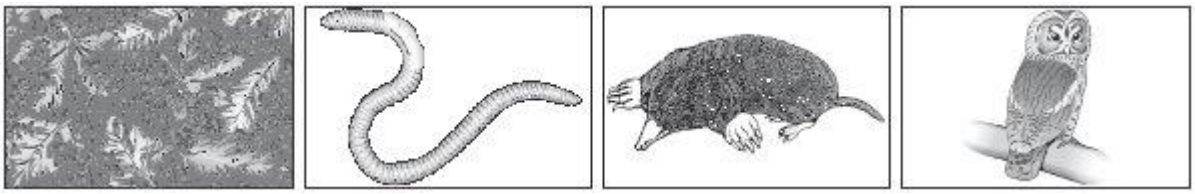
(a) Give **one** way a mole is suited for digging through soil.

.....

.....

1 mark

(b) Moles are part of the food chain shown below.



plant remains → worm → mole → owl

not to scale

(i) Which living thing in this food chain do moles eat?

.....

1 mark

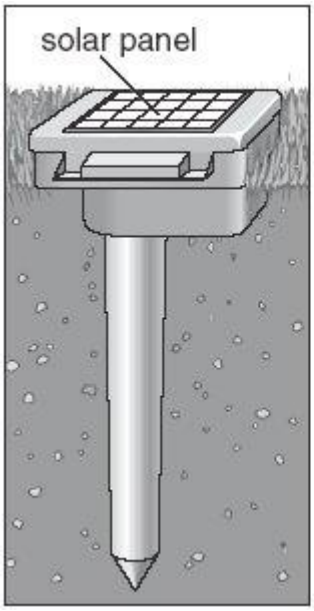
(ii) Which living thing in this food chain is a predator of moles?

.....

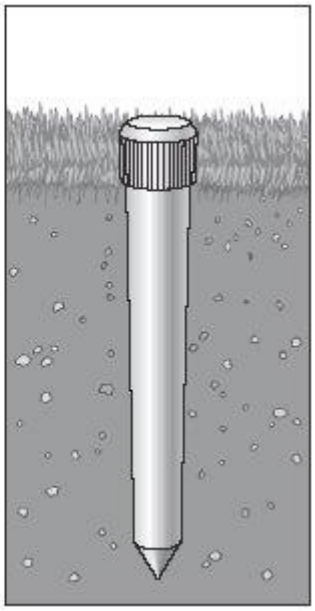
1 mark

(c) Some people use mole-scarers to get rid of moles from their gardens.

Two different mole-scarers are shown below.
They both produce sounds that scare moles away.



solar-powered mole-scarer



battery-powered mole-scarer

(i) Where does the energy come from for the solar-powered mole-scarer?

.....

1 mark

(ii) Suggest **one** reason for using a solar-powered mole-scarer instead of a battery-powered mole-scarer.

.....

.....

1 mark

(iii) Some gardeners use poison to kill moles.

Suggest **one** reason for using a mole-scarer rather than poison to get rid of moles.

.....

.....

1 mark
maximum 6 marks

Q1.

(a) any **one** from

- it has large **or** wide hands **or** paws
  accept 'its paws are like shovels'
  *accept 'big **or** strong feet'*

- it has a pointed head **or** face **or** nose
  accept 'it has a long nose'
  accept 'it is streamlined'

- it has strong **or** big claws
  *accept 'it has sharp **or** long claws **or** nails'*
  'claws' is insufficient
  *accept 'eyes protected by fur' **or** 'eyes deep in fur'*
  'it has small eyes' is insufficient

1 (L3)

(b) (i)
- worm

1 (L3)

(ii)
- owl
  accept 'bird'

1 (L3)

(c) (i)
- the Sun
  accept 'light'
  *'solar' **or** 'solar panel' **or** 'solar energy' are insufficient*

1 (L4)

(ii) any **one** from

- batteries have to be replaced
- solar power is free
- solar power does not run out
  accept 'it is a renewable source of power'
- it is cheaper to run
  accept 'it is cheaper'
  'it is environmentally friendly' is insufficient

1 (L4)

(iii) any **one** from

- it does not kill the moles
  accept 'it is humane'
  *'it is friendlier' **or** 'it is nicer' are insufficient*
- it does not harm other animals

accept 'it does not harm humans'

- poison kills **or** hurts moles
- poison kills **or** hurts owls **or** predators **or** other living things

  'poison is dangerous' is insufficient
- poison enters food chains

  'you do not have to get rid of dead moles'
  ***or** 'you can leave them in the tunnel' are insufficient*
  'you have to keep adding poison' is insufficient
  'so they do not become extinct' is insufficient

1 (L4)

[6]